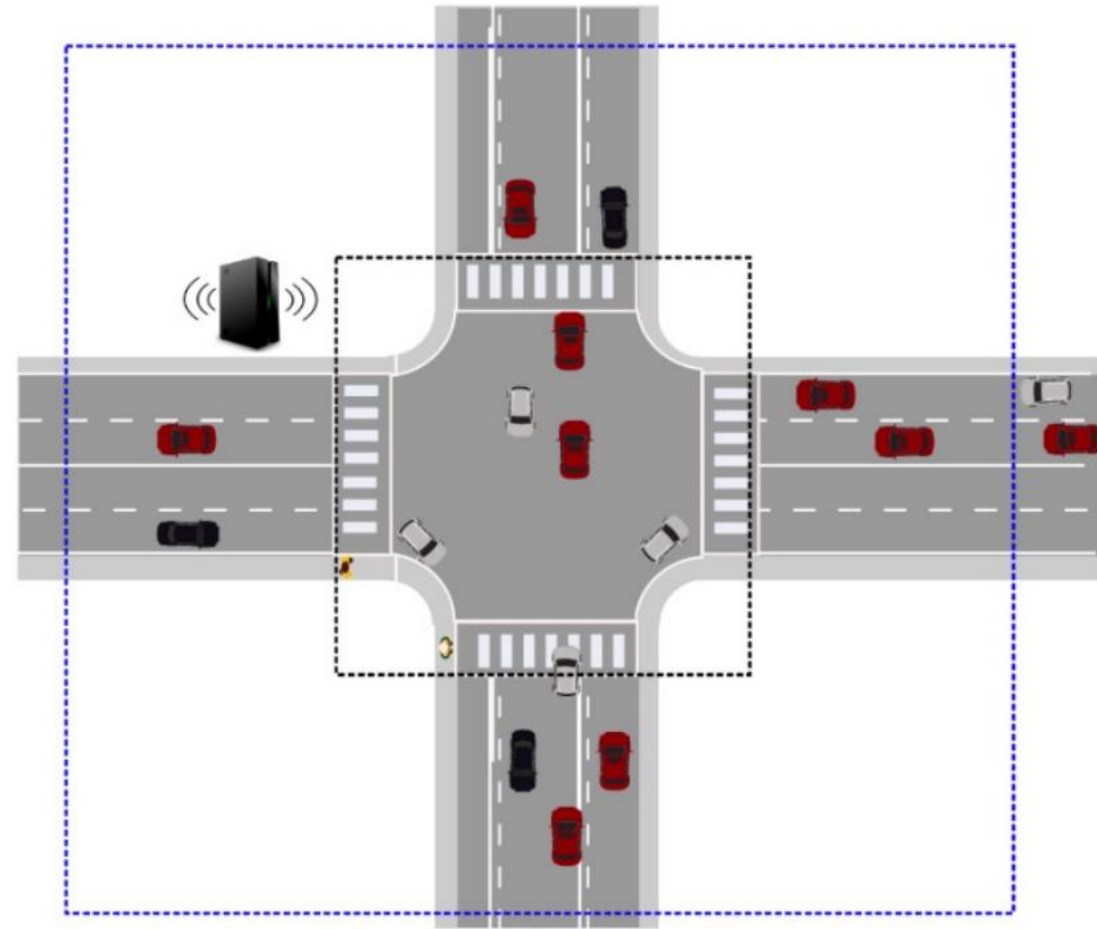
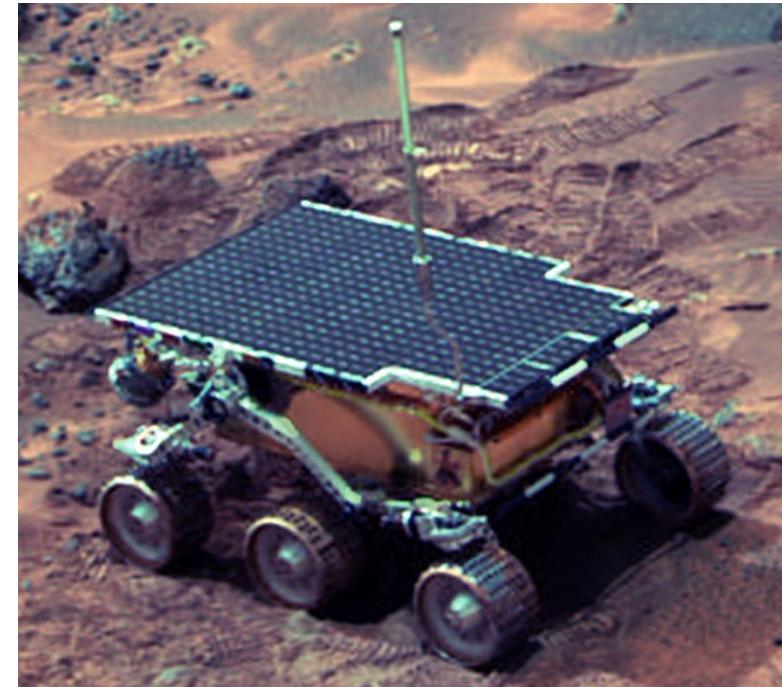
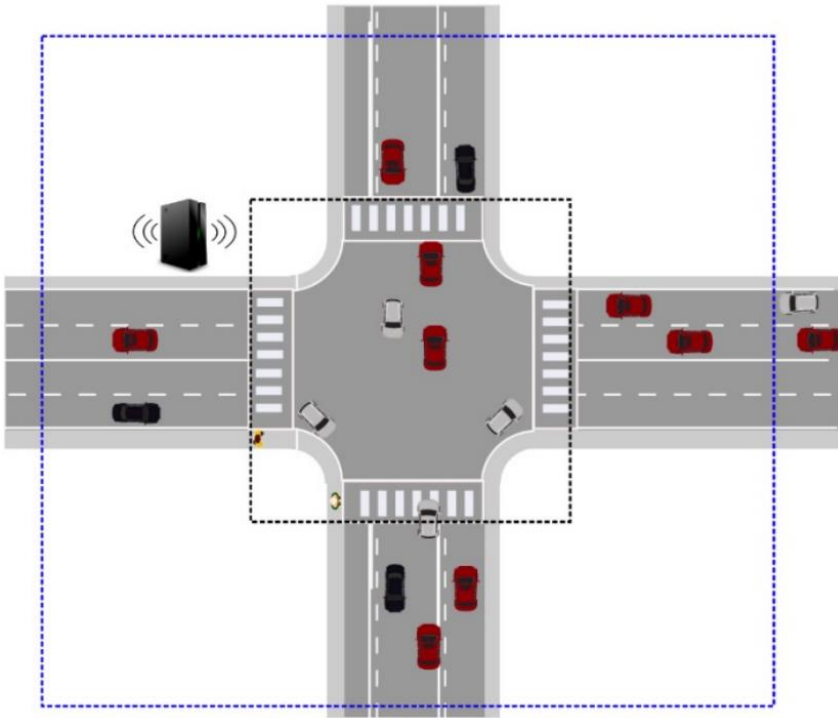


Resource Access Protocol In Autonomous Traffic Control System

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We are living in a world now, where data is everything and all processes of system shares the same data to maintain connectivity. The initial inspiration came while I was trying to solve the priority queue for a autonomous traffic control system. The problem was to setting the priority for each cars in the queue in the main shared resource. Similar problem can also be observed in Mars path finder incident happened in 1997.

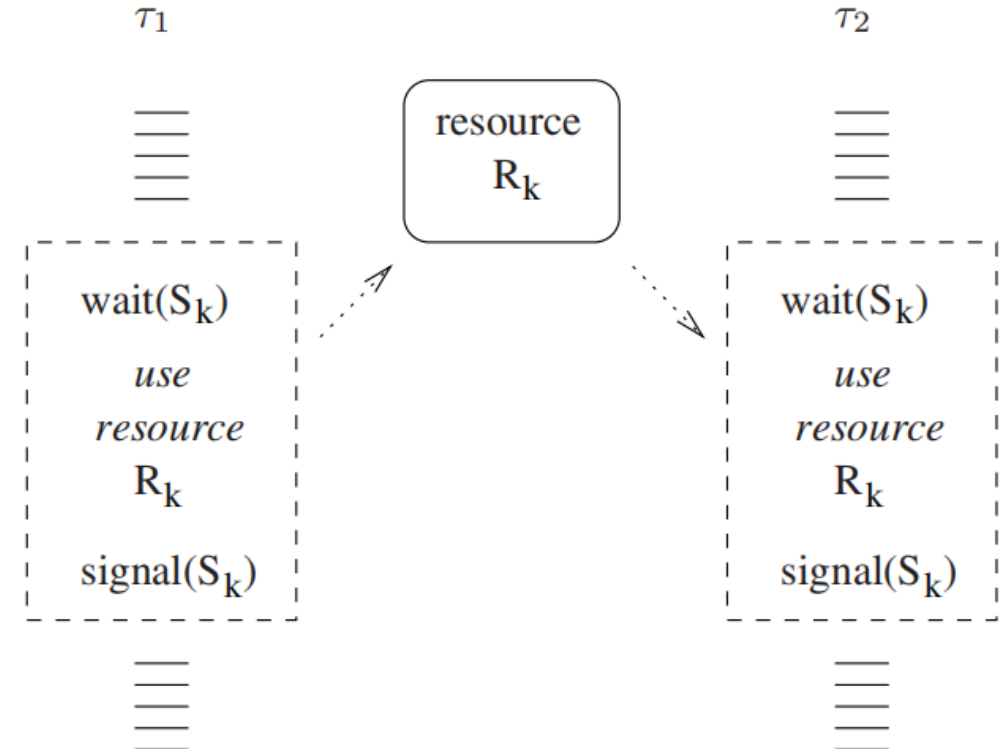


Shared Resources:

- Data structures,
- variables,
- main memory area,
- files,
- set of registers,
- I/O units,
- the processor, etc.

Mutual exclusion, critical Section

- When a job enters the critical section of a shared resource, the accesses to the shared resource from other jobs are blocked.

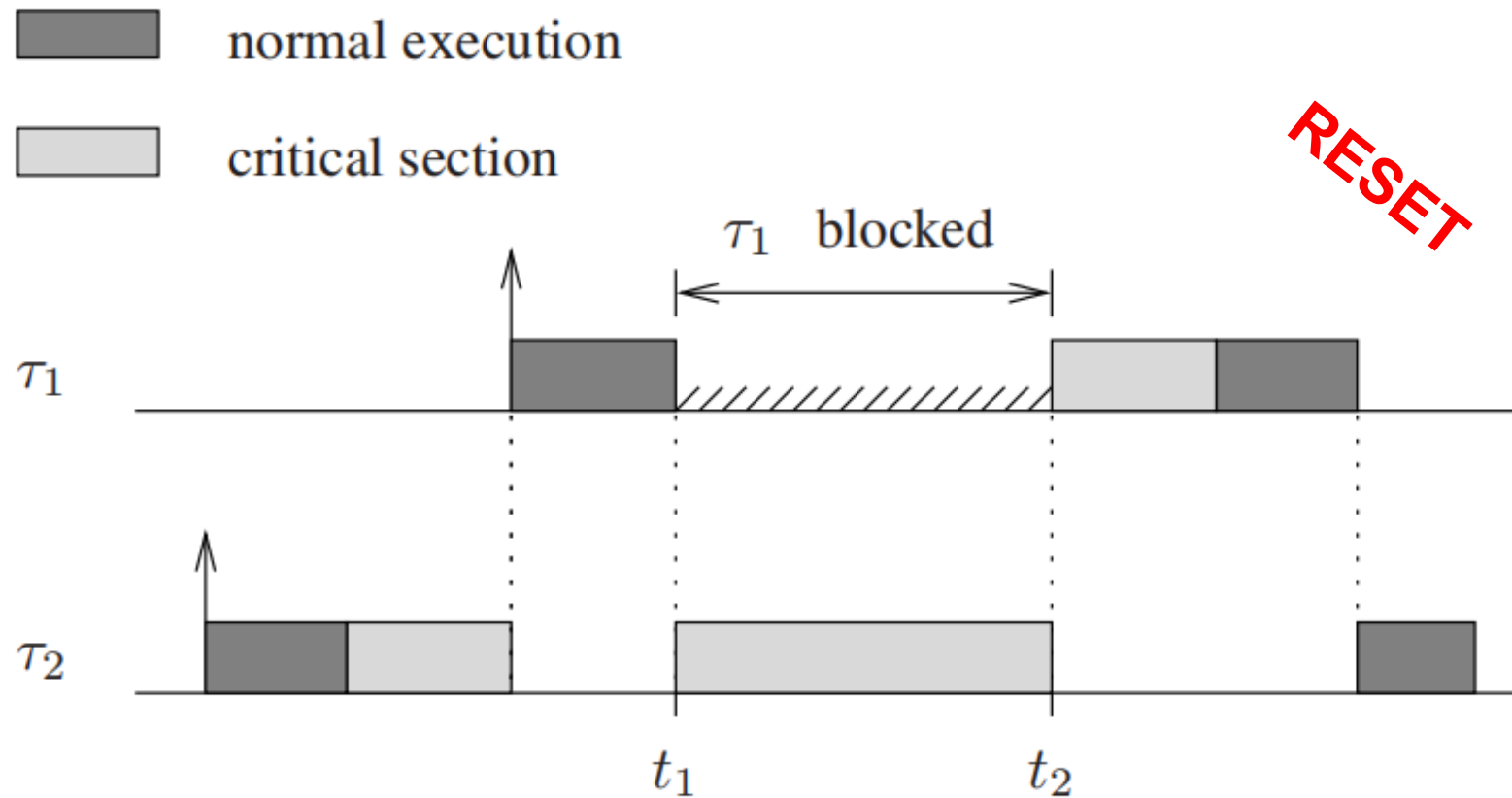


Server Queue

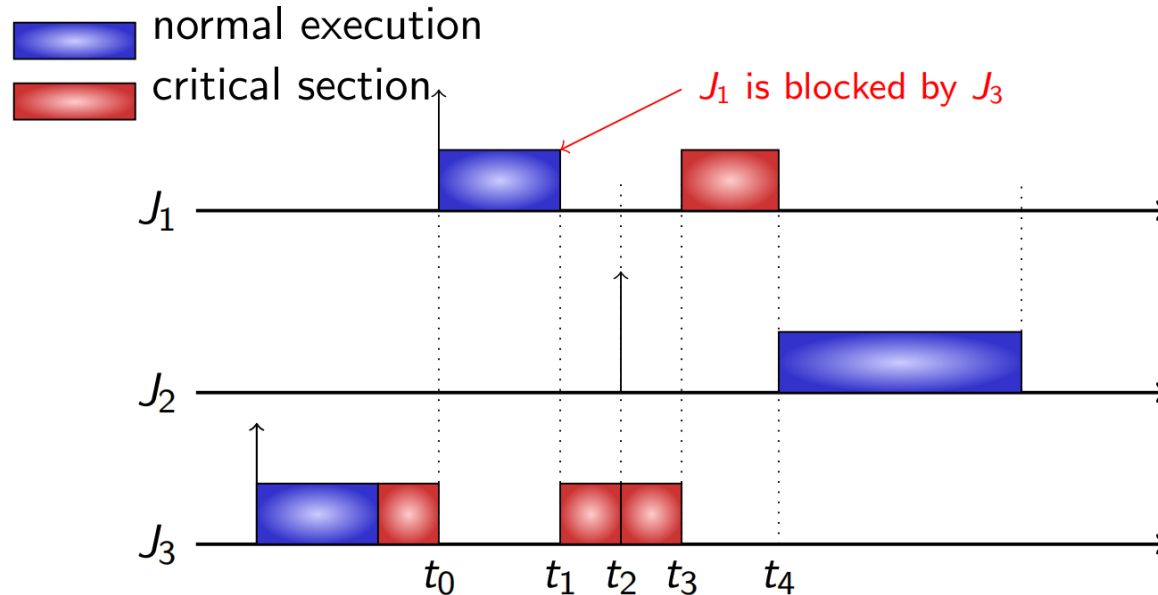
Priority 1	Car 1
Priority 2	Car 2
Priority 3	Car 3
Priority 4	Car 4

- In a 4-way cross sectional road, autonomous traffic control system is controlled by a priority server queue.
- Generally, vehicles are set to the queue as soon as they come.
- But fixed priority system for Resource Access creates a big problem.
- Like the Pathfinder system reset.

Priority Inversion Problem

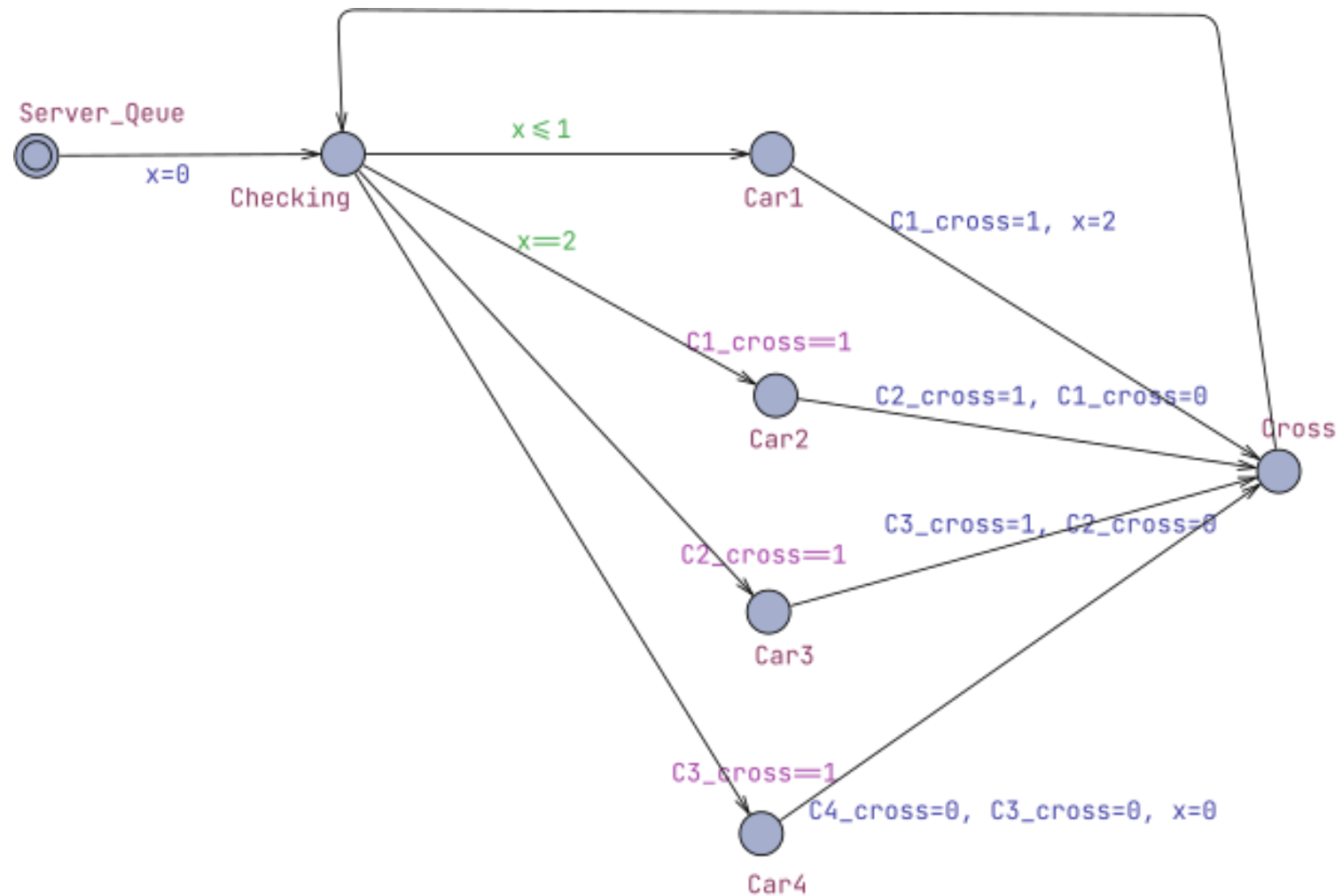


A higher priority job is *blocked* by a lower-priority job.

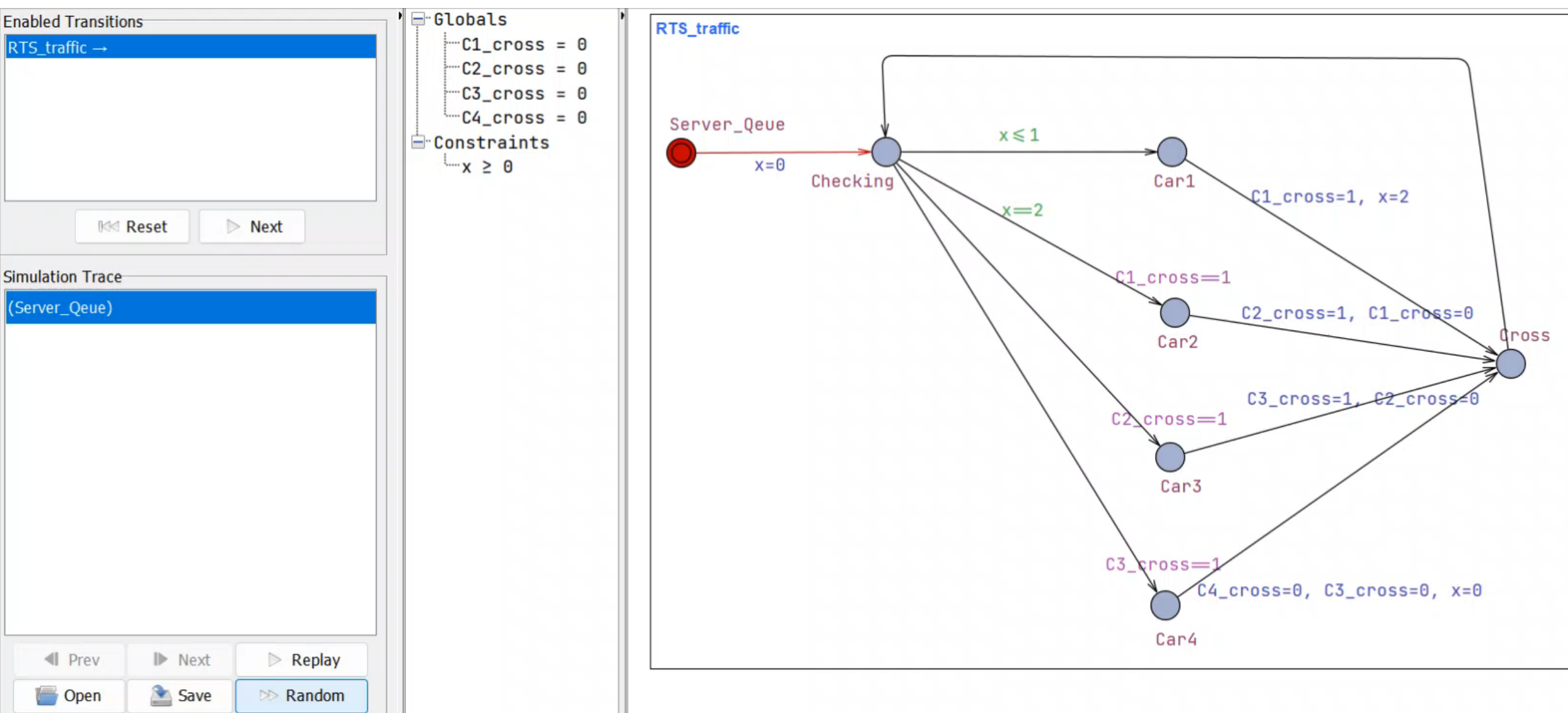


- T0: J_1 arrives and preempts J_3
- T1: J_3 inherits J_1 's priority
- T2: J_2 arrives and has a lower priority than J_3
- T3: J_3 leaves, and J_1 now preempts J_3 .
- T4: J_1 finishes, and J_2 is the highest-priority task.

When a lower-priority job blocks a higher-priority job, the priority of lower job is *promoted* to the priority level of highest-priority job.



UPPAL Simulation



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